

Flamenco: The Simple Open Source Render Farm

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ABSTRACT

Flamenco is the Open Source render farm software, developed by Blender Studio. It is aimed at performing tasks such as frame rendering and video encoding. This hands-on class will teach how to install and use it, and most importantly, how to adjust and extend it for your specific needs.

CCS CONCEPTS

• **Computing methodologies** → **Distributed computing methodologies**; **Rendering**; • **Computer systems organization** → **Client-server architectures**; • **Applied computing** → **Media arts**.

KEYWORDS

pipeline, render farm, distributed rendering, distributed computing

ACM Reference Format:

Sybren A. Stüvel. 2023. Flamenco: The Simple Open Source Render Farm. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Labs (SIGGRAPH '23 Labs)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588029.3599738>

1 INTRODUCTION

Flamenco is render farm software developed by Blender Studio. It is aimed at performing tasks, such as frame rendering and video encoding, either on a single machine or distributed across many. It supports any Blender version, hardware configuration and, albeit to a lesser degree, any other software with a command line interface.

The biggest design principles behind Flamenco are *simplicity* and *flexibility*. It is simple to set up, so that it can be used by people at home, but also by researchers on their lab computers, or studios on their own hardware. Being independent from any cloud infrastructure, and able to run fully offline, is vital for studios who need to protect their intellectual property.

Studio pipeline developers / TDs are able to customise the behaviour of Flamenco via an extensible "job types" system. In practice it means the addition of a single JavaScript file to the system. This kind of flexibility, coupled with ease of use (just unpack a ZIP and run), makes Flamenco quite unique in the space of distributed render management systems.

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SIGGRAPH '23 Labs, August 06–10, 2023, Los Angeles, CA, USA

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ACM ISBN 979-8-4007-0153-5/23/08.

<https://doi.org/10.1145/3588029.3599738>

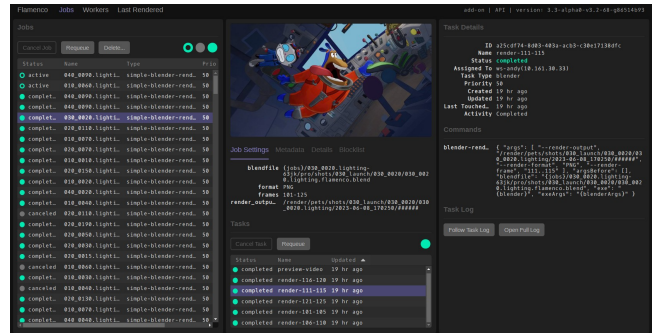


Figure 1: Flamenco in production at Blender Studio. © Blender Foundation | studio.blender.org

2 ATTENDEE EXPERIENCE

Attendees to be able to manage files (extract ZIP/TAR archives, create directories, create and edit plain text files). SIGGRAPH will provide a number of computers, but attendees can also work on their own laptops.

Experience with structured/code text files (YAML, JavaScript) is a plus. Having a code-oriented text editor (for example Visual Studio Code) available is recommended.

3 OUTLINE

The Hands-On Class will first teach how to install and use Flamenco. After that we will go over the possibilities to customize its behavior to specific needs. Finally some information will be given on the internal structure of Flamenco, and how to join this Open Source project.

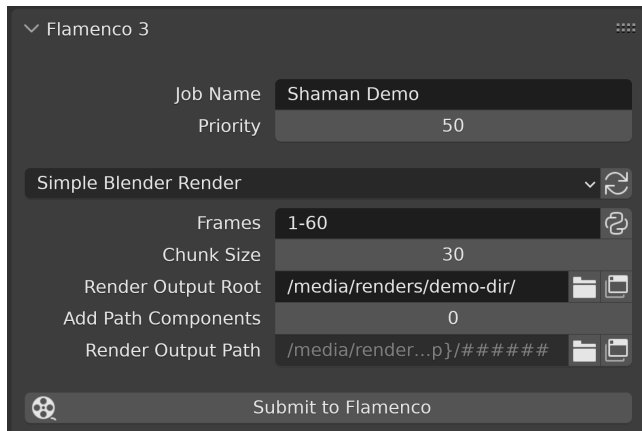
3.1 Introduction & Demo on own hardware

After an introduction of the Flamenco project and its authors, we will go over the steps to install and use Flamenco. During the demo, participants can download the latest version from <https://flamenco.blender.org/>. Finally participants will render a simple video on their own install.

3.2 Storage Options

We will go over Flamenco's various approaches to shared storage, which has a big impact on simplicity of use, the efficiency and speed of job submissions, and render job isolation.

- (1) Work directly on the shared storage
- (2) Create a copy for each render job
- (3) Shaman Storage System



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Job Name: Shaman Demo

Priority: 50

Simple Blender Render

Frames: 1-60

Chunk Size: 30

Render Output Root: /media/renders/demo-dir/

Add Path Components: 0

Render Output Path: /media/render...p}/#####

Submit to Flamenco

Figure 2: Flamenco job submission form in Blender

3.3 Writing a Custom Job Type

After an explanation of Flamenco’s job type system and data model, participants write a custom job type. For this we will take a built-in job type and expand it to include new metadata.

3.4 Cross-Platform Support

The class will conclude with a note about Flamenco’s cross platform support. Supporting a mixture of platforms, not only between the OS used by submitting artists, but also for the render nodes themselves, is an important factor in Flamenco’s success. However, by its very nature this is hard to experiment with this on a single computer, which is why this part will be limited to an explanation only.

3.5 Conclusion

The goal of this hands-on lab is to help people to start using Flamenco. They will become familiar with the file management strategies of Flamenco, its internal job structure, and customisation capabilities. They will also be provided with information on how to obtain further information and how to contribute to the project.